

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY



EMBEDDED SYSTEMS (23-203-0501)

ASSIGNMENT No. 2 (Mini Project using Microcontroller)

PROJECT TOPIC:

DESIGN AND IMPLEMENTATION OF 8051-BASED DIGITAL LOGIC GATE IC TESTER

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1 Abstract

This project presents the design and implementation of a low-cost microcontroller-based Digital Logic Gate IC Tester for verifying the functional operation of selected 74-series combinational logic ICs. The system uses an AT89S52 8051-family microcontroller as the central controller and a 14-pin ZIF socket for mounting the Device Under Test (DUT). Push buttons are used for user input, while a 16x2 LCD displays the selected IC and test status. The present implementation supports the 74LS00 NAND, 74LS02 NOR, and 74LS04 NOT gate ICs. The microcontroller applies the required input combinations, reads the corresponding outputs, compares them with the expected truth-table values stored in firmware, and indicates whether the IC passes or fails the functional test. The circuit is first validated through simulation and then implemented on perfboard. Firmware is developed for the 8051 using an Embedded C toolchain such as SDCC or Keil C51, with an Arduino Uno used as the programming interface for the AT89S52.

2 Introduction

2.1 Background

Digital logic ICs are fundamental building blocks used in digital electronics laboratories, embedded systems, and prototype circuits. A faulty IC can produce incorrect outputs and make debugging of an otherwise correct circuit difficult. Therefore, checking an IC before using it in a larger circuit is useful, particularly in an educational laboratory environment.

A digital IC tester automates this verification process by applying known input conditions to the Device Under Test (DUT), observing its output, and comparing the observed behaviour with the expected truth table. The microcontroller provides test stimuli, monitors the DUT response, and displays the final result.

2.2 Problem Statement

Manually testing individual logic ICs requires repeated wiring of inputs and observation of outputs. This becomes inconvenient when several ICs must be checked. Commercial IC testers may provide broader functionality but are not necessarily required for a basic laboratory application. Hence, the project aims to develop a compact and economical microcontroller-based tester for a small set of commonly used 74-series combinational logic ICs.

2.3 Objectives

The objectives of the project are:

- To design an 8051-based functional tester for selected 74-series digital logic ICs.
- To test 74LS08, 74LS32, 74LS00, 74LS02, 74LS04, and 74LS86 combinational logic ICs.
- To apply the required truth-table input combinations automatically.
- To compare the observed DUT outputs with expected logic values stored in firmware.
- To display the testing status and result on a 16x2 LCD.
- To provide simple user control using momentary push buttons.
- To validate the design using circuit simulation before hardware realization.

2.4 Scope

The current version is limited to selected combinational 74-series ICs, namely the 74LS08 (AND), 74LS32 (OR), 74LS00 (NAND), 74LS02 (NOR), 74LS04 (NOT), and 74LS86 (XOR). The hardware uses a 14-pin ZIF socket suitable for the selected devices. Sequential circuits such as flip-flops, counters, and shift registers are outside the present implementation and are considered under future scope.

3 System Design

3.1 Design Philosophy

The tester is designed around a simple functional-testing principle. Instead of measuring analog parameters of the IC, the system verifies its logical behaviour. For every supported IC, the firmware contains the expected input-output relationships. The AT89S52 generates the required input patterns, reads the DUT output pins, and compares the measured values with the expected truth-table values.

The design emphasizes low component count, simple user interaction, reconfigurability, and ease of construction. We use microcontroller, ZIF socket, LCD, and dedicated test logic as the basic architecture of a digital IC tester.

3.2 Working Principle

The basic operating sequence is:

1. Power is applied to the tester and the AT89S52 is initialized.
2. The 16x2 LCD displays the tester status and available test options.
3. The user places the DUT in the 14-pin ZIF socket.
4. Three push buttons (NEXT, SELECT, RESET) are used to select or initiate the required IC test.
5. The microcontroller configures the DUT input and output connections.
6. All required input combinations for the selected truth table are applied.
7. The corresponding DUT outputs are read by the microcontroller.
8. Each observed output is compared with the expected truth-table value.
9. The LCD displays the final test result.

3.3 Supported ICs

The firmware supports the six basic combinational logic functions: AND, OR, NAND, NOR, NOT, and XOR. Each IC is tested by applying all possible input combinations and comparing the observed output with the corresponding truth-table value; for the NOT gate, both input states are verified. The implemented test set covers the 74-series ICs listed in Table 1.

Table 1: Supported combinational ICs

IC	Logic Function	Gate Type
74LS08	AND	Quad 2-input
74LS32	OR	Quad 2-input
74LS00	NAND	Quad 2-input
74LS02	NOR	Quad 2-input
74LS04	NOT	Hex inverter
74LS86	XOR	Quad 2-input

3.4 Block Diagram

Figure 1 shows the proposed architecture. The AT89S52 is the central controller. The DUT is inserted into the 14-pin ZIF socket, push buttons provide user input, and the 16x2 LCD provides visual output. The power subsystem supplies the required regulated voltage to the circuit.

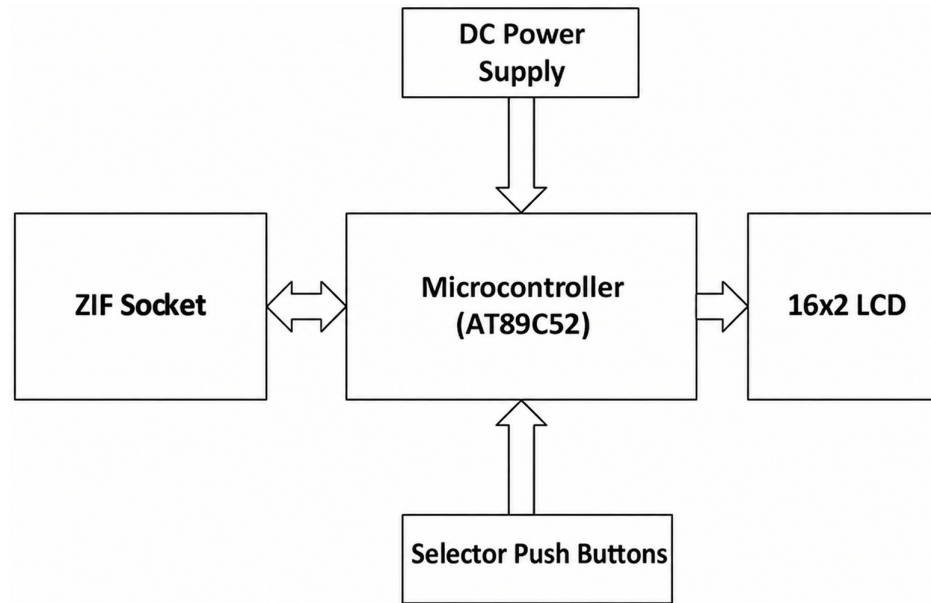


Figure 1: Block diagram of the Digital Logic Gate IC Tester

3.5 Flow Chart

The firmware flow begins with initialization, followed by user selection, truth-table execution, output comparison, and result display.

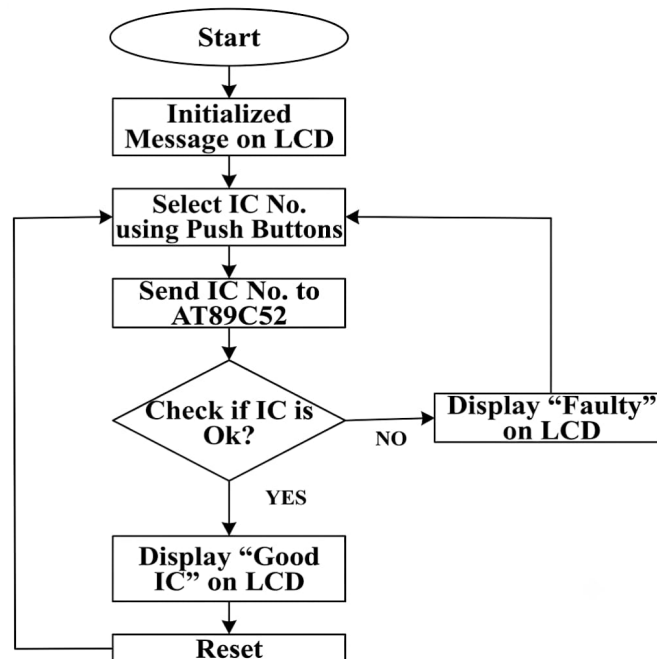


Figure 2: Firmware flow chart

4 Hardware

4.1 Major Hardware Components

The following table is based on the components purchased for the prototype. Prices are approximate estimates for budgeting and may vary with supplier and quantity.

Table 2: Purchased components, specifications, quantities, and estimated costs

Component	Specification	Qty.	Purpose	Est. Cost (INR)
AT89S52 MCU	8-bit 8051, DIP-40, 8 KB Flash, 256 B RAM	1	Main controller for test generation, response reading and result processing	100
DIP-40 IC socket	40-pin DIP socket	1	Removable socket for AT89S52	15
ZIF socket	14-pin ZIF	1	Holds the 14-pin DUT during testing	150
16x2 LCD	HD44780-compatible character LCD	1	Displays status, selection and test result	80
Crystal	12 MHz	1	AT89S52 clock source	10
Ceramic capacitors	22 pF	2	Crystal oscillator load capacitors	4
Resistors	1 k Ω	10	Biasing/current limiting/pull-up applications	10
Resistors	10 k Ω	2	Reset and bias circuitry	2
Potentiometer	10 k Ω	1	LCD contrast adjustment	10
Ceramic capacitors	100 nF	5	Local supply decoupling	5
Electrolytic capacitors	10 μ F	3	Supply filtering/stabilization	6
Tactile push buttons	Momentary, normally open	3	User input/control functions	6
ON/OFF switch	Small SPST type	1	Main power control	10
LEDs	Green	2	Status/power indication	6
LEDs	Red	2	Fault/status indication	6
Resistors	330 Ω	4–5	LED current limiting	5
Li-ion cell	18650	1	Portable power source	100
Battery holder	18650 holder	1	Battery connection	20
Charging module	TP4056 with protection	1	Li-ion charging and protection	25
Boost converter	MT3608, 5 V output	1	Generates regulated 5 V from Li-ion cell	30
Berg strips	Male-to-male	2	Breadboard/perfboard interconnections	20
Berg strips	Female-to-male	2	Interconnections/header access	20
Perfboard	General-purpose prototype board	2	Hardware assembly	40
Arduino UNO	ISP Interface	1	AVR USB ISP Programmer	-
74-Series ICs	Quad 2-Input AND, OR, NAND, NOR, NOT, XOR	6	DUT / supported test ICs	-
Approximate total				INR 744

4.2 AT89S52 Microcontroller

The AT89S52 is an 8-bit member of the 8051 family, providing the GPIO resources required to generate digital test inputs, read DUT outputs, interface with the LCD, and handle push-button inputs. Its pin configuration is shown in Figure 3. A 12 MHz crystal with two 22 pF capacitors is used for the oscillator circuit.

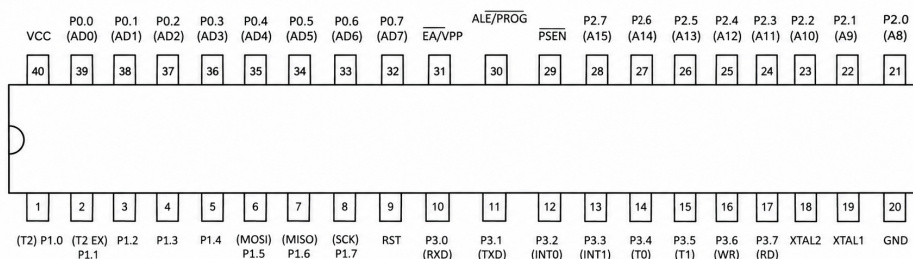


Figure 3: AT89S52 microcontroller pin configuration

4.3 14-pin ZIF Socket

The 14-pin Zero Insertion Force (ZIF) socket is the DUT interface. It allows the supported 14-pin logic ICs to be inserted and removed without repeatedly stressing their pins. The ZIF socket is particularly useful for a tester intended for repeated laboratory use.

4.4 16x2 LCD and Push Buttons

A 16x2 character LCD provides the user interface and displays the current operation and test result. Three momentary tactile push buttons (NEXT, SELECT, RESET) are used instead of matrix keypad for user control, reducing the interface hardware and simplifying the design.

4.5 Power Supply

The prototype uses an 18650 Li-ion cell with a TP4056 charging/protection module and an MT3608 boost converter. The boost converter provides the required 5 V supply for the 5-V logic system. Decoupling and filtering capacitors are distributed around the circuit to improve supply stability.

4.6 Indicators

Green and red LEDs are provided for status indication. Current-limiting 330 Ω resistors are used with the LEDs.

4.7 Circuit Simulation

Before hardware construction, the functional circuit is verified in a circuit simulation environment such as Proteus VSM. The simulation includes the AT89S52, oscillator/reset circuitry, LCD, push buttons, DUT interface, and representative supported logic IC.

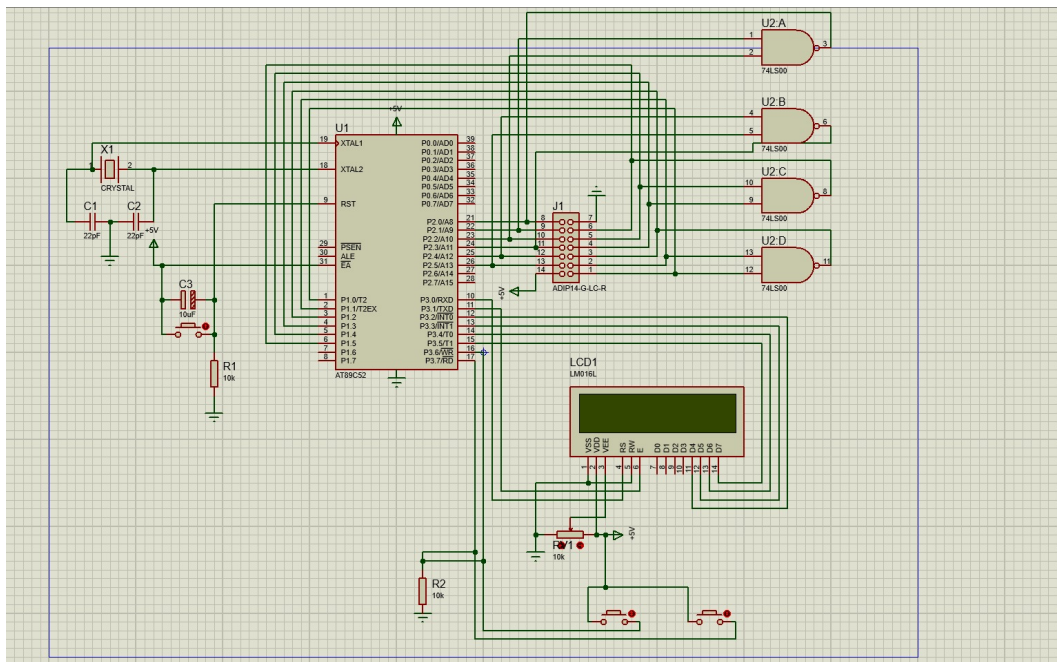


Figure 4: Circuit simulation of the Digital Logic Gate IC Tester

4.8 Simulation Output

The simulated LCD and status indicators are observed for representative good and faulty test cases. The expected output is obtained when the DUT responses match the stored truth-table values.

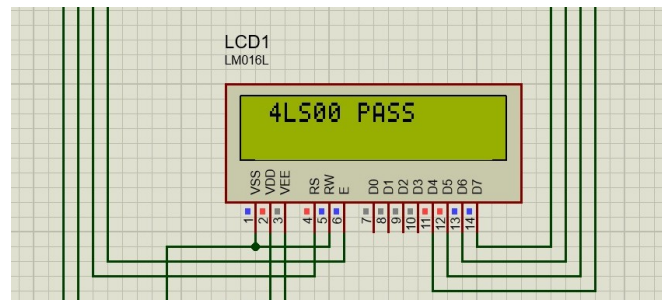


Figure 5: Representative simulation output

5 Software

5.1 Firmware Architecture

The firmware is divided into the following functional units:

- LCD initialization and display routines.
- Push-button input handling.
- DUT input/output pin configuration.
- Truth-table data for each supported IC.
- Test execution and output comparison.
- Good/Bad result generation and status indication.

The core testing algorithm is independent of the particular gate. The firmware selects the appropriate truth table, applies each input combination, reads the output, and declares the test successful only when all required combinations produce the expected results.

5.2 Firmware – Truth-Table Framework

Each supported gate type is represented in firmware by its own truth table, against which the DUT's observed output is checked row by row. Table 3 lays out the truth tables for all six supported gate types side by side, giving a compact, pseudo representation of the data the test routine consumes.

AND (74LS08)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR (74LS32)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT (74LS04)

A	Y
0	1
1	0

NAND (74LS00)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR (74LS02)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

XOR (74LS86)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Table 3: Truth tables of the six supported gate types, as stored in the firmware database

The test routine itself is gate-agnostic: it simply walks through whichever table above is selected,

applies each row's inputs to the DUT, and checks the DUT's output against that row's expected value. This is summarized in pseudocode below.

```

1 // Truth tables for all six gate types are pre-loaded at compile time
2 LOAD truth_table[AND, OR, NAND, NOR, NOT, XOR]
3
4 BEGIN test_routine(selected_gate):
5     table = truth_table[selected_gate]
6     result = GOOD
7
8     FOR EACH row IN table DO
9         SET DUT_input_A = row.A
10        SET DUT_input_B = row.B           // skipped for NOT (single input)
11        WAIT propagation_delay
12
13        READ DUT_output
14        IF DUT_output != row.expected THEN
15            result = BAD
16            BREAK
17        END IF
18    END FOR
19
20    DISPLAY result ON LCD
21    SET status_LED(result)               // green = GOOD, red = BAD
22    RETURN result
23 END

```

Listing 1: Pseudocode of the generic gate-testing algorithm

5.3 Arduino Uno as ISP Programmer

During development, an Arduino Uno is used as the ISP interface for programming the AT89S52 microcontroller. The Uno is configured with a suitable 8051 programming interface and connected to the AT89S52 through the required programming and control connections. The setup provides the necessary supply, common ground, clock, reset/programming-control, and data connections between the Arduino Uno and the target microcontroller.

A dedicated AVR USB ISP Programmer (USBasp) was also considered as an alternative programming interface. That USBasp-based programmer is primarily designed for in-system programming of AVR microcontrollers and is compatible with AVRDUDE. However, the present project uses the Arduino Uno instead of the USBasp programmer to interface with and program the AT89S52, which belongs to the 8051 family. Therefore, the Uno-based setup is treated as the programming interface for the AT89S52.

5.4 Software Development Tools

SDCC (Small Device C Compiler): Free, open-source, retargetable C compiler used to cross-compile the firmware for the 8051-family AT89S52 into an Intel HEX file.

Keil C51 / μ Vision: Commercial 8051 IDE with an integrated simulator/debugger, used where applicable to single-step and verify firmware logic before hardware testing.

Proteus VSM: Schematic capture and mixed-mode circuit simulation tool, used to co-simulate the compiled firmware against the full tester circuit before hardware realization.

Arduino IDE: Used only to flash the ArduinoISP sketch onto the Arduino Uno, turning it into an SPI-based ISP programmer.

avrdude: Used to flash the compiled Intel HEX firmware onto the AT89S52 via the Arduino Uno (acting

as an `stk500v1` ISP programmer), since `avrdude` includes native part support for the AT89S51/52/53 family over the shared AVR-style SPI ISP protocol.

6 Conclusion

The Digital Logic Gate IC Tester demonstrates the design of a compact embedded system for functional verification of selected 74-series combinational logic ICs. The AT89S52 performs the central test operation by applying predefined logic combinations to the DUT, reading its responses, and comparing them with the expected truth-table values. A 14-pin ZIF socket provides a convenient DUT interface, while the 16x2 LCD and push buttons provide a simple user interface.

6.1 Merits

- Low-cost implementation using readily available components.
- Automated truth-table-based functional testing.
- Simple user interface using push buttons and a 16x2 LCD.
- ZIF socket allows convenient insertion and removal of the DUT.
- Reconfigurable firmware allows additional supported ICs to be added.
- Portable operation is possible using the 18650 battery and boost converter.
- Useful as an educational laboratory tool for digital electronics.

6.2 Demerits

- The current version supports only a small number of combinational ICs.
- Sequential ICs such as flip-flops, counters, and shift registers are not currently supported.
- Functional testing does not characterize detailed electrical parameters such as propagation delay, noise margin, or maximum operating frequency.
- Manual perfboard construction can introduce wiring and assembly errors.
- The tester depends on correct DUT pin mapping and firmware definitions.

6.3 Future Scope

The primary future extension is to support sequential digital ICs. Unlike combinational logic, sequential circuits require state, clocking, reset conditions, and observation of outputs over time. The firmware and hardware can therefore be extended to test devices such as flip-flops, counters, and shift registers using appropriate timing sequences and state-transition tables.

Further possible extensions include:

- Expanding the supported IC database to additional 74-series devices.
- Adding dedicated protection for incorrect DUT insertion and abnormal conditions.
- Adding PC/UART connectivity for result logging and firmware/database updates.
- Improving the user interface and automatic IC identification.
- Developing a more robust PCB and enclosure for repeated laboratory use.

References

- [1] Atmel/Microchip, *AT89S52 8-bit Microcontroller Datasheet*.
- [2] Keil, *C51 Compiler and μ Vision Documentation*.
- [3] SDCC, *Small Device C Compiler Documentation*.